
Large Language Model Prediction Capabilities: Evidence from a Real-World Forecasting Tournament

Philipp Schoenegger
London School of Economics
and Political Science

Peter S. Park
MIT

Abstract

Accurately predicting the future would be an important milestone in the capabilities of artificial intelligence. However, research on the ability of large language models to provide probabilistic predictions about future events remains nascent. To empirically test this ability, we enrolled OpenAI’s state-of-the-art large language model, GPT-4, in a three-month forecasting tournament hosted on the Metaculus platform. The tournament, running from July to October 2023, attracted 843 participants and covered diverse topics including Big Tech, U.S. politics, viral outbreaks, and the Ukraine conflict. Focusing on binary forecasts, we show that GPT-4’s probabilistic forecasts are significantly less accurate than the median human-crowd forecasts. We find that GPT-4’s forecasts did not significantly differ from the no-information forecasting strategy of assigning a 50% probability to every question. We explore a potential explanation, that GPT-4 might be predisposed to predict probabilities close to the midpoint of the scale, but our data do not support this hypothesis. Overall, we find that GPT-4 significantly underperforms in real-world predictive tasks compared to median human-crowd forecasts. A potential explanation for this underperformance is that in real-world forecasting tournaments, the true answers are genuinely unknown at the time of prediction; unlike in other benchmark tasks like professional exams or time series forecasting, where strong performance may at least partly be due to the answers being memorized from the training data. This makes real-world forecasting tournaments an ideal environment for testing the generalized reasoning and prediction capabilities of artificial intelligence going forward.

1 Introduction

In the field of artificial intelligence (AI), large language models (LLMs) have recently shown surprising capabilities in a multitude of economically relevant tasks²⁵ that were previously thought to require human cognition.⁶ State-of-the-art LLMs are comprised of an extremely large amount of parameters—typically organized in terms of the Transformer architecture³⁷—and trained on a large corpus of Internet-based text data. This corpus of training data is used to train LLMs to predict the next sequence of tokens given an input. Despite the simplicity of the general task for which LLMs are trained—next token prediction—the resulting transfer learning causes LLMs to also become ostensibly proficient at a wide variety of specific tasks, including reading comprehension,^{30,38} summarization,¹¹ translation,¹⁵ coding,⁶ deception,³² medical-license exams^{6,27}, and bar exams.^{6,17}

The ostensibly impressive capabilities of LLMs come with several important caveats. One of the most important caveats to keep in mind is that when an LLM consistently outputs the true answers to questions in a task benchmark or exam, it is unclear whether the LLM’s true answers reflect a genuine understanding of the task: and as such, are likely to generalize to out-of-distribution settings.¹ An alternative hypothesis is that the training dataset contains at least part of the task benchmark’s questions and corresponding answers, and that the given LLM capability may thus not reflect a

genuine understanding that can generalize to analogous questions not in the training dataset.^{2,4,8,19} For example, consider a scenario where an organic-chemistry exam question is inputted into the LLM, and it outputs the true answer. It remains unclear whether this is due to the LLM possessing a deep understanding of the relevant organic-chemistry concepts, as opposed to simply reproducing the answer to the specific question contained in its training dataset. It is not trivial to rigorously formalize the dichotomy between genuine understanding and training-data memorization, as genuine understanding also ultimately arises from the relevant content in the training dataset. But the very real phenomenon of generalizability—or lack thereof—seems to be at the heart of the dichotomy.¹²

Given the problem of distinguishing whether an LLM’s proficiency at a task benchmark is due to genuine understanding or to memorizing the training data, we propose evaluating LLM capabilities in contexts where the true answers are unknown beforehand and as such cannot be part of the training data. An especially important such context is forecasting, a domain where the answers are initially unknown to everyone (even the human evaluators) until they are validated or falsified by future events.³³ We distinguish this context from the related but different forecasting context of zero-shot extrapolations of time series.¹³ In contrast, our definition of ‘forecasting’—producing accurate probabilistic predictions of future events—does not encounter the problem of distinguishing the benchmark data from the training data. As such, it allows for a more robust and generalizable evaluation of LLM capabilities: beyond the rote memorization of training data.

Our study evaluates the forecasting capabilities of GPT-4, a state-of-the-art LLM created by OpenAI,²⁸ by entering it in a forecasting tournament. Forecasting tournaments are competitions wherein individual forecasters provide probabilistic forecasts on questions concerning future events.³⁶ These predictions are then scored based on the accuracy of their forecasts: the closer one’s prediction is to the truth, the more likely one is to be rewarded. The collective accuracy of the predictions resulting from such forecasting tournaments hinges on the ‘wisdom of the crowd,’ i.e., the observation that aggregated forecasts of a group are often more accurate than the forecasts of individuals,^{7,20} even if judgments are correlated and biased.¹⁰ This setting allows us to test LLM capabilities in a context where the answer is genuinely unknown beforehand and thus cannot be contained within the training dataset. This setting may also be critical for whether LLMs are or will be able to rival—or even outperform—humans at jobs that require prescient decision-making.³¹ Accurately predicting the future is foundational to decision-making in most public and private sectors.³³ Consequently, comparing the forecasting capabilities of GPT-4 with those of humans promises to be a good test case for many potential economic applications of advanced AI.

As such, we have pre-registered—along with our analysis plans—the following null hypothesis:

The mean accuracy of GPT-4 forecasts is not different from the mean accuracy of median human-crowd forecasts. $H_0 : \mu_{\text{GPT-4}} = \mu_{\text{Human}}$

Human crowds in forecasting tournaments are among the best options for producing accurate probabilistic predictions.^{9,35} If a state-of-the-art LLM like GPT-4 is able to outperform a human crowd of forecasters, this would be consistent with the model having learned a deep understanding of the relevant capabilities, like probabilistic reasoning, generalization, and accurate prediction.

2 Methods

We conducted our study on Metaculus, a platform that hosts forecasting tournaments where members of the public can submit predictions, compete for prizes, and establish a forecasting track record. Metaculus’ forecasting platform has been employed in various academic and policy prediction contexts, such as the monkeypox outbreak in 2022²² and the COVID pandemic.²¹ We leveraged the opportunity provided by the launch of the Quarterly Cup²⁴ on July 3, 2023, as this three-month tournament offered an ideal context to evaluate LLM prediction capabilities. This setting provided us with a context where an LLM could compete with human forecasters to make probabilistic predictions about various topical questions.

Our questions were vetted by the Metaculus moderation team, who aimed to resolve the questions in a consistent and accurate manner in order to ensure high-quality data. The forecasting questions studied here were on a wide array of topics, such as U.S. industrial action disputes, military interventions in Niger, outbreaks of Marburg virus disease, and the Black Sea grain deal. For examples of the forecasting questions used in our study, see Table 1. For a full set of questions, see Appendix A.

Will the United Auto Workers call a strike against any of the Big Three Detroit automakers before September 19, 2023?
Will Mohamed Bazoum, Nigerien President, return to power before August 31, 2023?
Will India’s Chandrayaan-3 mission successfully land a rover on the moon?
Will the Black Sea grain deal be revived before October 1, 2023?
Will a non-proprietary LLM be in the top 5 of the chat.lmsys.org leaderboard on September 30, 2023?

Table 1: Examples of forecasting questions used in our study.

Our study’s questions were answered by both GPT-4 and a large set of human forecasters. The tournament concluded on October 4, with a total of 51 questions posed, and with 843 unique forecasters entering at least one prediction. For our analysis, we focused solely on the subset of binary questions, of which there were 23. This is because binary questions were the only types of questions that our LLM forecaster could straightforwardly answer without requiring additional human input. This could have biased the predictions, such as by drawing distributions based on quartile point estimates.

We used the web interface of GPT-4 at the default temperature value. Temperature is a hyperparameter that controls the randomness of a model’s output, with higher temperature settings resulting in more random outputs, and lower temperature settings resulting in more deterministic outputs. Our prompt was crafted by drawing on established research in forecasting, with the aim of steering GPT-4 towards comfortably providing numerical predictions and of enhancing the accuracy of these predictions. First, we prompted the model to emulate a superforecaster,³⁶ aligning with the best-practice recommendations of how to prompt models to act as if they were domain experts.³⁹ This part of the prompt enabled us to consistently avoid getting responses of a different format than explicit probabilistic forecasts.

Second, we grounded our prompt on research from the forecasting literature that highly complex qualitative rationales and the use of base rates are associated with forecasting accuracy.¹⁶ This part of the prompt was an attempt to not only elicit probabilistic forecasts, but to ensure that these are as highly accurate as possible.

The overall prompt is given by the following.

Prompt: In this chat, you are a superforecaster that has a strong track record of accurate forecasts of the future. As an experienced forecaster, you evaluate past data and trends carefully and aim to predict future events as accurately as you can, even though you cannot know the answer. This means you put probabilities on outcomes that you are uncertain about (ranging from 0 to 100%). When the outcome is continuous, you give me 25th interquartile ranges. You also quickly outline your rationale. In your rationales, you carefully consider the reasons for and against your probability estimate, you will make use of comparison classes of similar events and probabilities and take into account base rates and past events as well as other forecasts and predictions. You will also consider different perspectives.

Each question of the tournament was predicted via a single chat log, with the prompt positioned at the outset. Following this, GPT-4 was supplied with the background information, the specific resolution criteria of the question, and the final question text exactly as presented on Metaculus. This information, provided by Metaculus users or staff, helped mitigate the temporal distance between the training data set cutoff data and the question launch, which is less fundamental of an information gap for the human comparison group. For a full set of this information for one question, see Appendix B. For some questions, the median human-crowd prediction was available at the time of forecast; while for others, it was not. In the case of the latter, we included the median human-crowd prediction in the description provided to GPT-4.

Subsequently, we collected the probabilistic prediction and inputted it into the question, in percent form. In cases where the model output was a probability range, the mean probability of the two